Group 1 Test Plan

1. **Introduction**
    a. Test Plan Overview:
        i. This project is a Train Yard Management System. It manages how trains are built using engines and freight cars for food, wood, and oil.
    b. Test Plan Objective:
        i. The goal of testing is to make sure the system follows all safety rules and weight limits. We want to ensure the software correctly blocks any dangerous actions, like putting wood and oil cars next to each other.
2. **Scope**
    a. What we will test:
    b. Safety Rules: Engines must be at the front, and wood/oil cars cannot be neighbors
    c. Weight Limits: The train cannot be heavier than what the engines can pull (5,000 kg per engine) or the system limit of 20,000 kg
    d. Car Management: Adding and removing cars correctly
    e. What we will not test:
    f. The demo main function provided at the start (it is only for show)
    g. Physical train hardware or tracks.
3. **Test Strategy**
    a. System Test (Blackbox)
        i. Objective: Verify that the system follows all business rules without looking at the internal code.
        ii. Focus: Testing the Lead Rule (engines at the front), Weight Constraints (20,000 kg limit), and the Adjacency Rule (no WOOD next to OIL).
        iii. Deliverable: An Excel document containing at least six functions with four sets of test data each.
    b. Automated Test (Whitebox & Git Hooks)
        i. Unit Testing: Individual functions will be tested for internal logic paths and branch coverage using the Visual Studio test framework.
        ii. Git Hook Automation: A pre-push script will be implemented to automatically run unit tests before any code is shared to the GitHub repository.
        iii. Requirement: If any test fails, the push is aborted to prevent bugs from entering the main branch.
    c. Integration Test
        i. Objective: Examine how multiple functions interact to complete complex tasks rather than testing them in isolation.
        ii. Example: Passing the results of a weight calculation into an engine power check, then into a car removal function.
        iii. Requirement: Each case must use at least three unique data inputs and involve at least three unique functions.
    d. User Acceptance Test
        i. Objective: The final validation phase to prove the system meets the original business requirements for the "client".

        ii. Requirement: Create exactly one specific test case for every project requirement listed in the documentation.

4. **Environmental Requirements**
5. **Hardware Requirements**
    a. Workstations: Testing is primarily conducted on team member's local development machines (workstations)
    b. Architecture: Since the project involves a provided C-based codebase with TrainCar structures, the hardware must support a standard C compiler (eg. x86_64 or AMR64)
6. Software Requirements
    a. Operating System: Windows, macOS, or Linux
    b. Version Control: Github is mandatory for storing all code, documentation, and test plans
    c. Project Management: Jira is required for tracking all tasks and bugs via a Kanban board
    d. Development Environment: A C compiler (GCC or Clang) and an IDE/Text Editor (VS Code, Visual Studio).
7. Testing Tools & Harness
    a. Test Harness: We are responsible to build a harness to execute Blackbox(ms3), whitebox(ms4) and integration (ms5) tests
    b. Git Hook Automation: For MS4 we must implement a pre-push Git hook that automatically runs unit tests before code is pushed to GitHub
    c. Traceability: We must maintain a Requirements Traceability Matrix(RTM) to map every test case to the system's safety and weight requirements (eg the lead rule or adjacency rule
8. Environmental Setup for Execution
    a. Weekly Synchronization: We must meet at least once per week to synchronize testing efforts and project goals.
    b. Branch Management: All code must eventually be merged into the main/master branch for final delivery in MS6
9. **Execution Strategy**
10. **Test Schedule**
    a. During Milestone 2 we will mainly focus on preparing the test plan and understanding what parts of the system need to be tested. At this stage our team will review the project requirements and the starter code that was provided. Based on this, we will identify the main rules of the system such as engine placement, weight limits, and safety conditions between different car types.
    b. Our team will start creating test scenarios that can later be used to verify if the system behaves correctly. For example we will consider situations where cars are added or removed, or when the total train weight changes. We will also define what the expected result should be in those situations so that later we can compare the actual behavior with the expected one.
    c. Most of the work in this milestone is focused on planning the testing process rather than executing the tests. We expect the preparation of the test plan, test scenarios and documentation to be completed during the MS2 period. By the end of this

milestone our team should have a clear idea of how the system will be tested and what conditions must be validated during testing.

11. **Control Procedures**
    a. Reviews
        i. During the project our team will regularly review the work that has been done, especially the test plan and related documentation. We will check if the planned tests actually cover the important requirements of the system such as engine rules, weight limits and safety constraints. These reviews help us make sure that nothing important is missing in our testing plan and that the testing approach still matches the project requirements.
    b. Bug Review Meetings
        i. If we notice possible issues or mistakes while preparing the test plan or analyzing the system requirements, our team will discuss them during our meetings. In these discussions we will try to understand why the issue happened and what the correct behavior of the system should be. After that we decide what needs to be fixed or clarified before continuing with the project work.
    c. Change Request
        i. Sometimes while analyzing the system or preparing the test plan we might realize that some changes are needed. If this happens our team will discuss the change and decide if it should be applied. Before making the change we will also think about how it could affect the system design or the testing process.
    d. Defect Reporting
        i. If we identify any defects or problems during the testing preparation or later testing activities, they will be documented and tracked. Our team will record what the issue is, how it appeared and what the expected system behavior should be. These issues will be shared with the team so that they can be fixed and verified again later.
12. **Functions To Be Tested**
13. **Resources and Responsibilities**
14. **Deliverables**
    a. **MS1**
    b. **MS2**
    c. **MS3**
    d. **MS4**
    e. **MS5**
    f. **MS6**
15. **Suspension / Exit Criteria**
16. **Resumption Criteria**
17. **Dependencies**
18. **Risks**
19. **Tools**
20. **Documentation**
21. **Approvals**